HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and integrated **Python**-based AI agents and agentic workflows into production software solutions, handling approximately 500 requests per minute.
- Designed and built **RAG**-based solutions for enterprise data, integrating with vector databases to enable efficient information retrieval for AI applications.
- Orchestrated complex AI capabilities using **LangGraph** and **LangChain**, managing state and execution flow for multi-step agentic processes.
- Authored **Java Spring Boot** microservices to provide RESTful APIs for internal tooling, processing over 1,000 transactions daily through a robust data validation layer.
- Refactored existing **Python** ETL scripts to improve performance and reliability, reducing data processing times by an average of **30%** for critical reporting tasks.
- Containerized AI applications and services using **Docker**, deploying them to an **AWS EC2** environment managed via Jenkins pipelines for automated build and deployment.
- Collaborated with cross-functional teams to define technical requirements and estimate effort for new feature development within an Agile Scrum framework.

SKILLS

AI/ML – Agentic AI, RAG, LangGraph, LangChain, MCP, LLMs, Vector Databases, Model Integration

Languages – Python, Java, C#

Backend Frameworks – Spring Boot, FastAPI, Flask

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle, Vector DBs

Methodologies – Agile, Scrum, CI/CD

PROJECTS

AI Agent Orchestration for Enterprise Workflows

- Developed a Python application utilizing **LangChain** and **LangGraph** to orchestrate a sequence of AI agents for automating complex business processes, processing over 2,000 workflow instances.
- Integrated a **RAG** component with an internal knowledge base to provide context-aware responses, improving the accuracy of agent-generated outputs by an estimated **15%**.
- Containerized the agent application using **Docker** and deployed it to an **AWS Lambda** environment for scalable execution, ensuring high availability for critical workflows.

Python-based Data Pipeline for AI Feature Engineering

- Engineered a **Python** data pipeline on **AWS S3** and **EC2** to extract, transform, and load data from disparate sources for AI model training, handling datasets up to **500GB**.
- Implemented data validation and cleansing steps using **pandas** and **NumPy**, reducing data quality issues in downstream AI feature sets by approximately **20%**.
- Automated pipeline execution using scheduled jobs within the **Linux** environment, ensuring timely availability of processed data for machine learning teams.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)